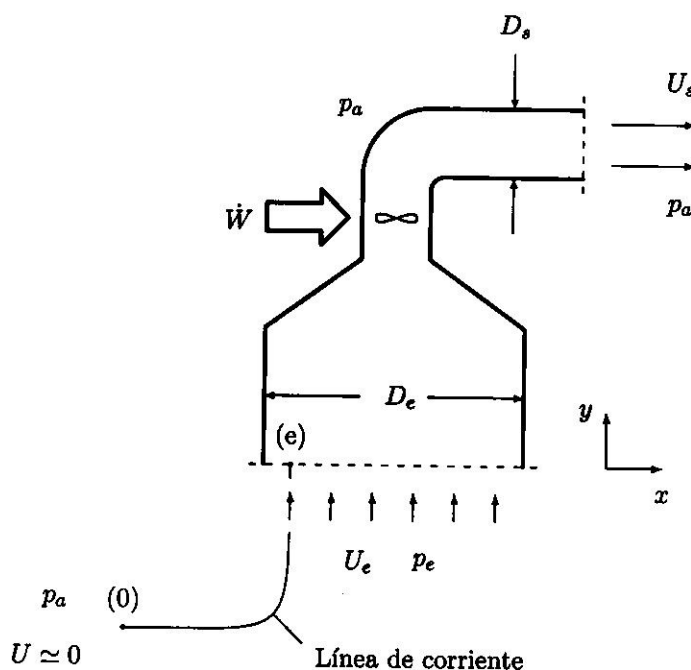


CL10

The ventilation device shown in the figure takes air out of a room at pressure p_a through a circular nozzle of diameter D_e . The air is discharged to the open atmosphere, also at pressure p_a , through a pipe of diameter $D_s < D_e$. To force the air motion the device includes a fan, which exerts a rate of work per unit time $\dot{W}$ on the air stream. To simplify the solution you may assume that the air density ρ and viscosity μ are constant and that gravity has a negligible effect.

1. Using the continuity equation, obtain a relationship between the values of the velocity at the entrance and exit sections U_e and U_s , assumed to be uniform.
2. Make use of Bernoulli's equation upstream from the entrance section (i.e., along a stream line between (0) and (e)) to obtain an equation for the pressure and velocity at the entrance section p_e and U_e .
3. Determine the two components of the force F_x and F_y acting on the device. Include in the computation the contribution of the stagnant surrounding air.
4. Use the energy equation to compute the value of U_s as a function of the power $\dot{W}$. Assume that the air temperature has the same values at the entrance and exit sections and that the device is thermally insulated.
5. Obtain the numerical values of U_s , U_e , p_e , F_x and F_y for $D_e = 2$ m, $D_s = 0,1$ m, $\rho = 1,2$ kg/m³, $\mu = 1,8 \times 10^{-5}$ kg/(m.s), $\dot{W} = 1000$ Watt, $p_a = 101300$ Pa.

Note: the following quantities are known: D_e , D_s , ρ , μ , $\dot{W}$, p_a .



$$\textcircled{1} U_e \frac{\pi D_e^2}{4} = U_s \frac{\pi D_s^2}{4} \quad \textcircled{2} p_a = p_e + \frac{1}{2} \rho U_e^2 \quad \textcircled{3} -\rho U_e^2 \pi \frac{D_e^2}{4} \bar{e}_y + \rho U_s^2 \pi \frac{D_s^2}{4} \bar{e}_x = (p_e - p_a) \pi \frac{D_e^2}{4} \bar{e}_y - \bar{F}$$

$$F_x = -\rho U_s^2 \pi \frac{D_s^2}{4}, \quad F_y = \rho U_e^2 \pi \frac{D_e^2}{4} + (p_e - p_a) \pi \frac{D_e^2}{4}$$

$$\textcircled{4} \rho U_s \pi \frac{D_s^2}{4} \left[\left(\frac{p_e}{\rho} + \frac{U_s^2}{2} \right) - \left(\frac{p_e}{\rho} + \frac{U_e^2}{2} \right) \right] = p_e U_e \pi \frac{D_e^2}{4} - p_s U_s \pi \frac{D_s^2}{4} + \dot{W} \Rightarrow \dot{W} = \rho \pi \frac{D_s^2}{4} U_s \left[\left(\frac{p_a}{\rho} + \frac{U_s^2}{2} \right) - \left(\frac{p_a}{\rho} + \frac{U_e^2}{2} \right) \right] \Rightarrow$$

$$\dot{W} = \rho \pi \frac{D_s^2}{8} U_s^3$$